NNN-F-001643 (DSA-DM)
March 1. 1971
Interim revision of
Fed. Spec. GG-F-591b
March 27, 1957

INTERIM FEDERAL SPECIFICATION

FORCEPS, MICROSCOPE COVER GLASS

This interim Federal Specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-F-591.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers forceps suitable for handling microscope slides and cover glasses.

1.2 Classification.

1.2.1 Types. Forceps shall be of the following types as specified (see 6.2).

Type I	Mood Design.
Type II	Microscope Cover Glass.
Type III	Gooseneck.
Type IV	Sharp Points, Curved.

FSC 6640

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited).
QQ-N-290	Nickel Plating (Electrodeposited).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Federal

Fed. Std. 123	Marking for Domestic Shipment (Civilian Agencies).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B-164	Nickel-Copper Alloy Rod and Bar.
E-18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

(Applications for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 Materials.

3.1.1 Carbon steel. Carbon steel shall conform to the requirements of either AISI 1020, 1022, or 1023 carbon steel strip. The carbon steel shall have a Rockwell Hardness of B75-B100 when tested as specified in 4.4.1.

3.1.2 Spring wire. Spring wire shall conform to the requirements of AISI 1040 carbon steel, hard drawn, steel wire.

3.1.3 Monel metal. Monel metal shall conform to the requirements of ASTM Designation B-164.

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3.2 Style and design.

3.2.1 Type I. Shall be fabricated of a suitable one piece carbon or corrosion-resisting steel spring wire, heavily nickel plated. Shall be 5 ± 1/4 inch in over-all length. Shall have a loop, approximately 3/4 inch in diameter, at the closed end, one finger grip, approximately 5/8 inch in diameter, near the center length of each arm, and an integral guide loop approximately 5/16 x 1 inch in outside dimensions near the open end. At the extreme open end, there shall be two jaw tips, the upper square cut measuring .018 inch in diameter and the lower formed into an oblong shape approximately 5/16 x 5/16 inch in outside dimensions. Shall have an oblong shaped rest approximately 1-7/16 x 5/16 inch in outside dimensions. located approximately 5/8 inch below the lower jaw tip. **Type I forceps shall conform to figure 1 for style and design.**

3.2.2 Type II. Type II forceps shall be straight-tipped cover glass spring type forceps. Forceps shall be fabricated from carbon steel as specified in 3.1.1. Forceps shall be of two-piece construction with the closed end being resistance-welded. The welded end shall be smoothly finished. Forceps shall have thin, flat, beveled or slightly tapered tips, and the finger grips shall be corrugated. Forceps shall be provided with a guide pin located near the tips. The overall length shall be 4-1/2 ± 1/8 in. and the maximum width shall be 5/16 ± 1/16 in. The width of the tips shall be 1/8 ± 1/32 in. Type II forceps shall conform to figure 2 for style and design.

3.2.3 Type III. Type III forceps shall be the Gooseneck design. Forceps shall be fabricated from Monel metal as specified in 3.1.3. Forceps shall be of a two-piece construction with the closed end being resistance-welded. The welded end shall be smoothly finished. The jaws shall be parallel and curved at the tips. Inside of the tips shall be serrated. Forceps shall have an over-all length of 5-3/4 ± 1/8 in. and the maximum width of the finger grip shall be 5/16 ± 1/16 in. The prong portion of the forceps which extends from the finger grips to the jaws shall be a nominal 1/8 in. thick. Type III forceps shall conform to figure 3 for style and design.

3.2.4 Type IV. Type IV forceps shall be a curved, sharp points type forceps. Forceps shall be fabricated from carbon steel as specified in 3.1.1. Forceps shall be of a two-piece construction with the closed end being resistance welded. The welded end shall be smoothly finished. The jaws shall be parallel and curved at the tips. Inside of the tips and the finger grips on the handles shall be serrated. The overall length of the forceps shall be 4-1/2 ± 1/8 in. and the maximum width of the closed end shall be 5/16 ± 1/16 in. Type IV forceps shall conform to figure 4 for style and design.

3.3 Finish.

3.3.1 Plating. Forceps fabricated from carbon steel sheet and carbon steel wire shall have a plating of chromium applied over a plating of nickel over copper. The chromium plating shall be in accordance with class 1, type 1 of QQ-C-320 and the nickel plating shall be in accordance with class 1, type III of QQ-N-290.

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3.3.2 Monel metal. Forceps fabricated from monel metal shall have a bright finish.

3.4 Resistance to corrosion. Forceps covered by this specification that are plated shall show no evidence of corrosion when tested as specified in 4.4.2.

3.5 Identification marking. Except for type I, forceps shall be legibly and permanently marked with the name or registered trademark of the manufacturer.

3.6 Workmanship. Workmanship shall be first class throughout. Forceps shall be free of sharp edges, burs, and scratches and other defects which detract from their appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the carbon steel sheet, carbon steel wire and monel metal, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the lot number, the lot size, hardness, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be one pair of forceps.

Table I. Sampling for Examination

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	2.5
Monor defects	II	4.0
For dimensional examination	S-2	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, except that the acceptance number shall be zero for all sample sizes. Unit of product for testing purposes shall be one pair of forceps.

4.3 Examination. Forceps shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with table II. The forceps shall be examined for defects but shall not be restricted to the classified possible defects.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Table II. Categories and defects

Major	
101	Plating not present, not evenly deposited or not deposited over entire forceps, where required.
102	Forceps not free from distortion.
103	Forceps not free from sharp edges.
104	Forceps not free from corrosion.
105	Guide pin binds not aligned with hole or missing, where required.
Minor	
201	Forceps not free from grease, dirt or other foreign matter.
202	Name or registered trademark of manufacturer not legibly or permanently marked or missing.

4.4 Sampling for tests. Tests shall be conducted to determine compliance with specification requirements. Unit of product for test purposes shall be one pair of forceps. The acceptable quality level shall be zero.

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Unit of product	Characteristic	Requirement	Test procedure	Inspection level
One forceps	Rockwell hardness	3.1.1	4.4.1	S-2
One forceps	Resistance to corrosion	3.4	4.4.2	S-2

4.4.1 Rockwell hardness test. Rockwell hardness shall be determined in accordance with ASTM Designation E-18.

4.4.2 Boil test. Before testing, the forceps shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the forceps shall be boiled in distilled water in a large glass vessel, for 30 minutes and allowed to remain submerged for an additional 24 hours.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One forceps, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	360 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be over-packed in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566, PPP-B-636, or PPP-B-676. Commercial colors will be acceptable when carton conforming to PPP-B-566 or PPP-B-676 is utilized. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Forceps covered by this specification are intended to handle and hold microscope slides and cover glasses during laboratory staining procedures.

6.2 Ordering data. Procurement documents should specify the following:

(a) **Title, number and date of this specification.**
(b) Specify type required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Federal Stock Number (FSN).

6.3 This specification does not cover all types of the commodity indicated by the title of this specification, or those which are commercially **available,** but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number - 6640-0640

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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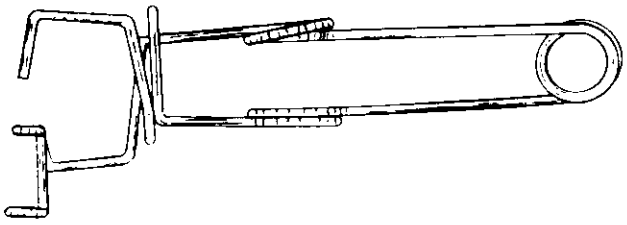


FIGURE 1



FIGURE 2

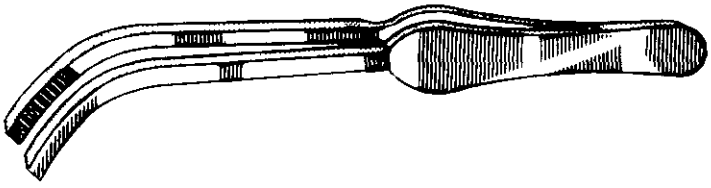


FIGURE 3



FIGURE 4

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION
NNN-F-001643

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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